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Undecidability Part - 1

Complete Course on Theory of Computation for GATE CSIT

Ankit Kumar • Lesson 78 • Oct 3, 2026

CS & IT Engineering



Lecture Number



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Topics to be Covered

1. GATE PYQ DCFL & CFL

1. $L = \{a^n b^n \mid n \geq 1$
2. $L = \{a^n b^n \mid n \geq 0$
3. $L = \{a^n b^{2n} \mid n \geq 1$
4. $L = \{a^n b^{3n} \mid n \geq 1$
5. $L = \{a^n b^{kn} \mid n \geq 1$
6. $L = \{a^{2n} b^n \mid n \geq 1$
7. $L = \{a^{3n} b^n \mid n \geq 1$
8. $L = \{a^{kn} b^n \mid n \geq 1$
9. $L = \{a^{2n} b^{3n} \mid n \geq 1$
10. $L = \{a^{3n} b^{5n} \mid n \geq 1$
11. $L = \{a^{xn} b^{yn} \mid n \geq 1$

12. $L = \{a^{n+1} b^{2n} \mid n \geq 0$

13. $L = \{a^{n} b^{n+1} \mid n \geq 1$

14. $L = \{a^{n} b^{n+2} \mid n \geq 1$

15. $L = \{a^{n+1} b^{n} \mid n \geq 1$

16. $L = \{a^{n+2} b^{n} \mid n \geq 1$

17. $L = \{a^{m} b^{n} \mid m > n$

18. $L = \{a^{m} b^{n} \mid m \geq n$

19. $L = \{a^{n} b^{m} \mid m \geq n+3$

20. $L = \{a^{m} b^{n} \mid m < n$

21. $L = \{a^{m} b^{n} \mid m \leq n$

22. $L = \{a^{m} b^{n} \mid m = n$

23. $L = \{a^{m} b^{n} \mid m \neq n$

24. $L = \{a^{m} b^{n} \mid \ m > n \text{ or } m < n$

25. $L = \{a^n b^n \mid n \geq 0\} \cup \{a\}$
26. $L = \{a^n b^n \mid n \geq 0\} \cup \{b\}$
27. $L = \{a^n b^n \mid n \geq 0\} \cup \{a^n \mid n \geq 0\}$
28. $L = \{a^n b^n \mid n \geq 0\} \cup \{b^n \mid n \geq 0\}$
29. $L = a^n c b^n \mid n \geq 1$
30. $L = a^m b^m c^n \mid m, n \geq 1$
31. $L = a^m b^{m+2} c^n \mid m, n \geq 1$
32. $L = a^m b^n c^n \mid m, n \geq 1$
33. $L = a^m b^n c^m \mid m, n \geq 1$
34. $L = a^m b^n c^p \mid m=n$
35. $L = a^m b^n c^p \mid n=p$
36. $L = a^m b^n c^p \mid m=p$
37. $L = a^n b^m c^{m+n} \mid m, n \geq 1$
38. $L = a^{n+2} b^m c^{m+n} \mid m, n \geq 1$

39. $L = a^{m+n}\, b^m\, c^n \mid m, n \geq 0$

40. $L = a^m\, b^{m+n}\, c^n \mid m, n \geq 0$

41. $L = a^n\, b^n\, c^m\, d^m \mid m, n \geq 0$

42. $L = a^n\, b^m\, c^m\, d^n \mid m, n \geq 0$

43. $L = a^n\, b^m\, c^n\, d^m \mid m, n \geq 1$

44. $L = a^m\, b^n\, c^{m-n} \mid m,n \geq 0$

45. $L = a^{m-n}\, b^m\, c^n \mid m,n \geq 0$

46. $L = a^m\, b^{m-n}\, c^n \mid m,n \geq 0$

47. $L = a^{m+n}\, b^{n+k}\, c^{k+m} \mid m,n\, k \geq 0$

48. $L = a^{m+n}\, b^{m+n}\, c^{m+n} \mid m,n \geq 0$

49. $L = a^{m+n}\, b^{m+n}\, c^m \mid m,n \geq 0$

50. $L = \{a^n\, b^n \mid n \geq 0\} \cup \{a^n\, b^{2n} \mid n \geq 0\}$

51. $L = \{c\, a^n b^n \mid n \geq 1\} \cup \{d a^n b^{2n} \mid n \geq 1\}$

52. $L = \{a^n b^n \mid n \geq 0\} \cup \{a^n b^{n+1} \mid n \geq 0\}$

53. $L = \{a^n b^n \mid n \geq 0\} \cup \{a^n b^{n+3} \mid n \geq 0\}$

54. $L = \{a^n b^m \mid m=n \text{ or } m=n+1$

55. $L = \{a^n b^m \mid m=n \text{ or } m=n+1 \text{ or } m=n+3$

56. $L = \{a^n b^m \mid m=n \text{ or } m=2n$

57. $L = \{a^m b^n c^p \mid m=n \text{ or } n=p$

58. $L = \{a^m b^n c^p \mid m=p \text{ or } n=p$

59. $L = \{a^m b^n c^p \mid m=p \text{ or } m=n$

60. $L = a^m b^n c^p d^q \mid m+n = p+q \text{ , } m, n, p, q \geq 1$

61. $L = a^m b^n c^p d^q \mid m+q = n+p \text{ , } m, n, p, q \geq 1$

62. $L = a^m b^n c^p d^q \mid m+p = n+q \text{ , } m, n, p, q \geq 1$

63. $L = a^m b^n c^p d^q \mid m-n=p-q$

64. $L = a^m b^n c^p d^q \mid m-q=n-p$

65. $L = a^m b^n c^p d^q \mid m-p=n-q$

66. $L = \{ww^R \mid w \in (a^*)$

67. $L = \{ww^R \mid w \in (a, b)^*$ or Even Length Palindrome

68. $L = wxw^R \mid w \in (a, b)^*, X \in (a, b)$ or Odd Length Palindrome

69. $L = wxw^R \mid w \in (a, b)^*, X \in (a, b, \epsilon)$ or All Palindrome

70. $L = wcw^R \mid w \in (a, b)^*$

71. $L = wc^m w^R \mid w \in (a, b)^*, m \geq 1$

72. $L = wc^m w^R \mid w \in (a, b)^*, m \geq 0$

73. $L = w\#w^R \mid w \in (a, b)^*$

74. $L = wcw^R \mid w \in (a, b, c)^*$

75. $L = wc^m w^R \mid w \in (a, b, c)^*, m \geq 1$

76. $L = w\#w^R \mid w \in (a, b, \#)^*$

77. $L = \{w \mid w \in (a, b)^* \text{ and } n_a(w) = n_b(w)$

78. $L = \{w \mid w \in (a, b)^* \text{ and } n_a(w) > n_b(w)$

79. $L = \{w \mid w \in (a, b)^* \text{ and } n_a(w) < n_b(w)$

80. $L = \{w \mid w \in (a, b)^* \text{ and } n_a(w) \neq n_b(w)$

81. $L = \{w \mid w \in \sum^* \mid W \text{ is a string of balanced parenthesis over } \sum = \{ (,) \}$

82. $L = \{w\, a^n\, b^n \mid w \in (a, b)^*\ n \geq 0\}$=Regular ⟶

83. $L = \{w\, a^n\, b^n \mid w \in (a, b)^*\ n \geq 1\}$

84. $L = \{ww^R\, a^n\, b^n \mid w \in (a, b)^*\ n \geq 0\}$

85. $L = \{w\#w^R \#\, a^n \#\, b^n \mid w \in (a, b)^*\ n \geq 0\}$

86. $L = \{ww^R \#\, a^n\, b^n \mid w \in (a, b)^*\ n \geq 0\}$

87. $L = \{w\#w^R \# a^n b^n \mid w \in (a, b)^* \; n \geq 0\}$

88. $L = \{ww^R \#w \mid w \in (a, b)^* \}$

89. $L = \{w\#w^R \# w\#w^R \mid w \in (a, b)^* \}$

90. $L = \{w\# a^n b^n \mid w \in (a, b)^* \; n \geq 1\}$

91. $L = \{w\#w^R \# x \mid w, x \in (a, b)^* \}$

92. $L = \{w_1\#{w_1}^R \# w_2\#{w_2}^R \mid w_1, w_2 \in (a, b)^* \}$

93. $L = \{wa^n b^n w^R \mid w \in \{a, b\}^*, n \geq 0\}$

94. $L = \{a^n w b^n \mid w \in \{a, b\}^*, n \geq 0\}$

95. $L = \{wa^n w^R b^n \mid w \in \{a, b\}^*, n \geq 0\}$

96. $L = \{wxw^R \mid w, x \in \{a, b\}^*\}$

97. $L = \{ww^R xx^R \mid w, x \in \{a, b\}^*\}$

98. $L = \{wxx^R w^R \mid w, x \in \{a, b\}^*\}$

99. $L = \{wxw^R x^R \mid w, x \in \{a, b\}^*\}$

100. $L = \{a^n b^n \mid n \geq 0\} \cup \{a^n b^{3n} \mid n \geq 0 \cup \{a^n b^{5n} \mid n \geq 0\}$

82. $L = \{w\, a^n b^n \mid w \in (a, b)^*\ n \geq 0\}$ = Regular

$\eta = 0 \longrightarrow (a+b)^* \cdot \epsilon \cdot \epsilon = (a+b)^*$

83. $L = \{w\, a^n b^n \mid w \in (a, b)^*\ n \geq 1\}$

SKIP Push Pop

Nondeterminism

$\longrightarrow$ CFL But Not DCFL

84. $L = \{ww^R\, a^n b^n \mid w \in (a, b)^*\ n \geq 0\}$

DPDA Can't Find middle so of CFL But Not DCFL

85. $L = \{w\#w^R\,\#\,a^n\,\#\,b^n \mid w \in (a, b)^*\ n \geq 0\}$ $\longrightarrow$ DCFL But Not RL

Push SKIP Pop SKIP Push SKIP Pop

83. $(a+b)^* ab$

$(a+b)^* aabb$

$(a+b)^* aacabbb$

86. $L = \{ww^R \# a^n b^n \mid w \in (a, b)^* \; n \geq 0\}$ ⟶ CFL But Not DCFL

DPDA Can't Find middle

87. $L = \{w\#w^R \# a^n b^n \mid w \in (a, b)^* \; n \geq 0\}$ ⟶ DCFL But Not RL



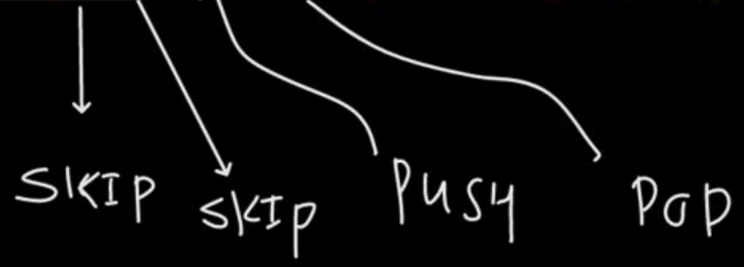
88. $L = \{ww^R \#w \mid w \in (a, b)^* \}$ ⟶ CSL But Not CFL



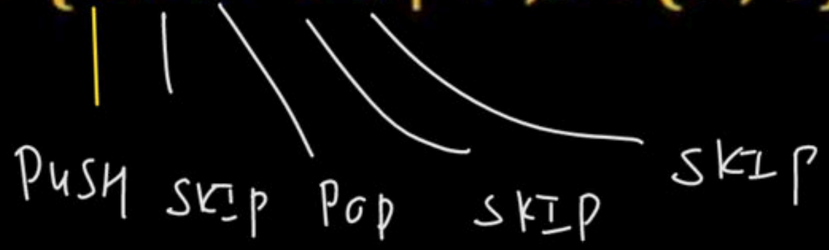
89. $L = \{w\#w^R \# w\#w^R \mid w \in (a, b)^* \}$ ⟶ CSL But Not CFL



90. $L = \{w\# a^n b^n \mid w \in (a, b)^*\ n \geq 1\}$ ⟶ DCFL But Not RL



91. $L = \{w\#w^R\#x \mid w, x \in (a, b)^*\}$ ⟶ DCFL But Not RL



92. $L = \{w_1\#w_1^R\#w_2\#w_2^R \mid w_1, w_2 \in (a, b)^*\}$ ⟶ DCFL But Not RL



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93. $L = \{wa^n b^n w^R | w \in \{a, b\}^*, n \geq 0\}$



⟶ CFL But Not DCFL

DPDA can't Find middle element But NPDA can Find middle element

CFGs Have recursive structure

1. $ww^R \mid w \in (a,b)^*$ OR even Length Palindrome

abba , baab , abaab a

Palindrome

a ~~~ a , b ~~~ b

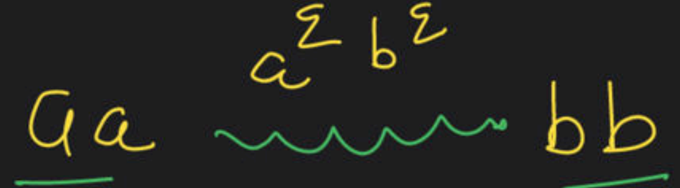
Palindrome

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$L = a^n b^n / n \geq 0$

$L = \{ab, aabb, aaabbb, aaaabbbb, \ldots\}$



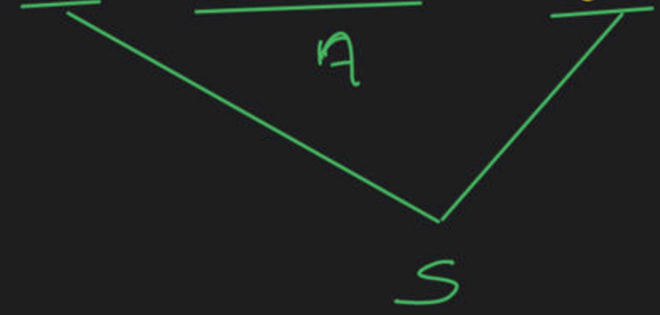


Important Language

① ww^R OR even Length Palindrome

$$S \rightarrow aSa \mid bSb \mid \in$$

② $w\ a^n\ b^n\ w^R \longrightarrow$ CFL

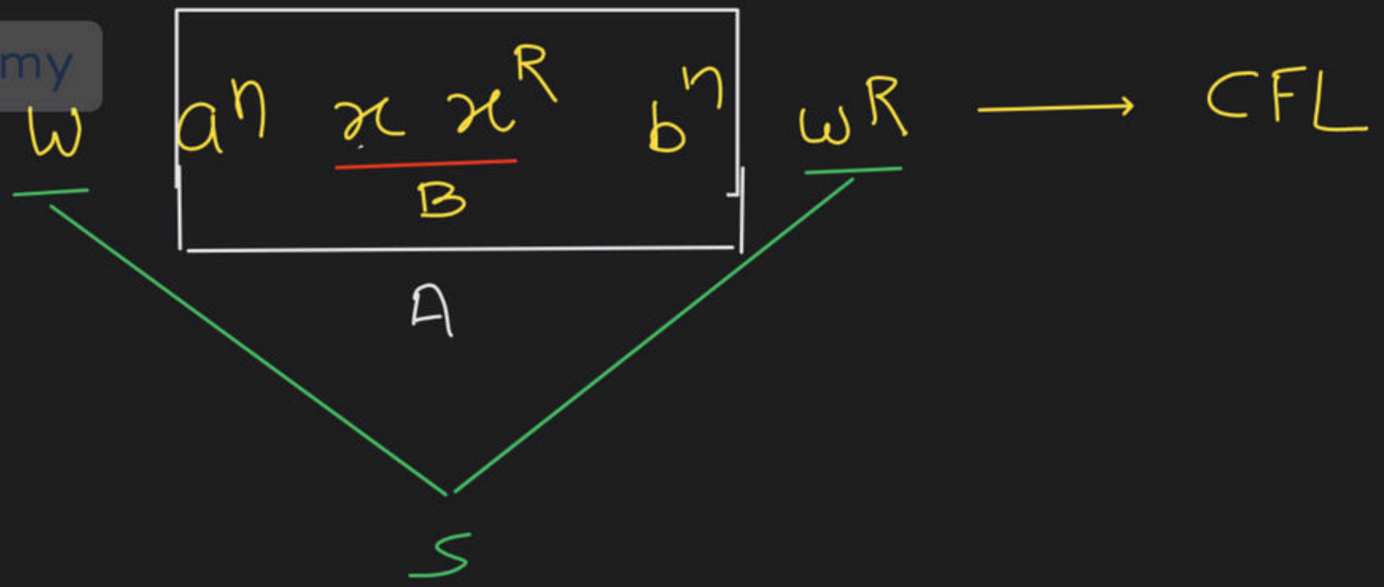


$$S \rightarrow aSa \mid bSb \mid A$$

$$A \rightarrow aAb \mid \in$$

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③



w^R ⟶ CFL

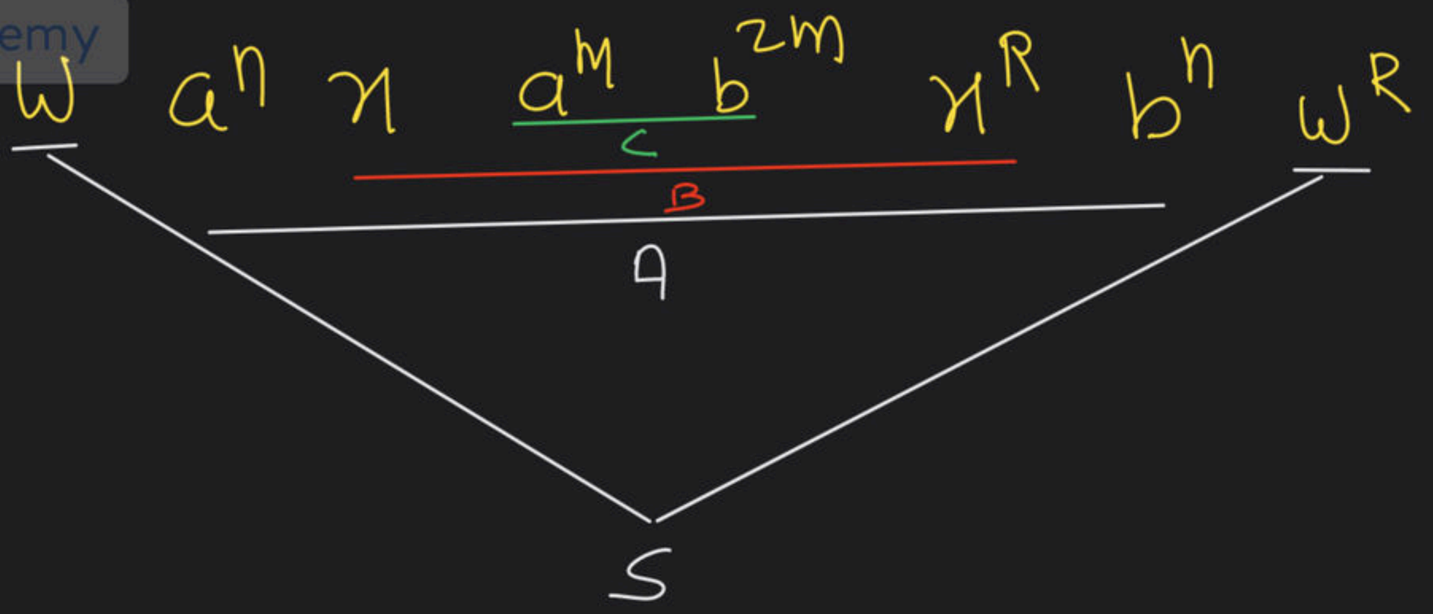
$S \longrightarrow aSa \mid bSb \mid A$

$A \longrightarrow aAb \mid B$

$B \longrightarrow aBa \mid bBb \mid \epsilon$

④ $w\ a^n\ x\ a^m\ b^{2m}\ x^R\ b^n\ w^R \longrightarrow$ CFL

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$S \longrightarrow aSa \mid bSb \mid A$

$A \longrightarrow aAb \mid B$

$B \longrightarrow aBa \mid bBa \mid C$

$C \longrightarrow aCbb \mid \epsilon$

94. $L = \{a^n w b^n | w \in \{a, b\}^*, n \geq 0\}$ RL

Put $n=0 \longrightarrow a^0 (a+b)^* b^0 = (a+b)^* \longrightarrow$ RL

$n=3$, $w = abb$

$w = aaaaaabbb$

$n=0$

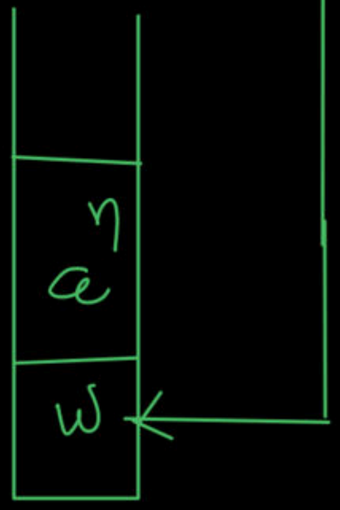
$\underbrace{aaa}_{a^3}\ \underbrace{abb}_{w}\ \underbrace{bbb}_{b^3} \longrightarrow a^0 \underline{aaa\ abb\ bbb}\ b^0$

95. $L = \{wa^n w^R b^n \mid w \in \{a,b\}^*, n \geq 0\}$ ⟶ CSL But Not CFL



$L = \quad w \; a^n \; w^R \; b^n$

PUSH PUSH



We can't compare with Bottom of stack so it is Not CFL
it is CSL

96. $L = \{ w a^n w^R b^n \mid w \in (a,b)^*, n \geq 0 \}$

(i) If we put $n = 0 \longrightarrow w w^R$

$L = \{ w w^R \}$

$x = \underbrace{bbbb}_{w} \underbrace{aaaa}_{a^n} \underbrace{bbbb}_{w^R} \underbrace{bbbb}_{b^n}$ this string

Belong to the Lang $L = w a^n w^R b^n$ But

can't be generated by ww^R

ww^R can't count a's and b's

96. $L = \{ w a^n w^R b^n \mid w \in (a,b)^*, n \geq 0 \}$

If we put $w = \epsilon \longrightarrow \epsilon \cdot a^n \epsilon \cdot b^n \rightarrow a^n b^n$

$L = a^n b^n$

$x = \underbrace{bbbb}_{w}\ \underbrace{aaaa}_{a^n}\ \underbrace{bbbb}_{w^R}\ \underbrace{bbbb}_{b^n}$ this string

Belong to the Laug $L = w a^n w^R b^n$ But

can't be generated by $a^n b^n$

$a^n b^n$ can't count w and w^R

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96. $L = \{wxw^R \mid w, x \in \{a, b\}^*\}$ ⟶ Regular

$w = \in \longrightarrow \in \cdot (a+b)^* \cdot \in = (a+b)^* \longrightarrow RL$

97. $L = \{ww^R xx^R | w, x \in \{a, b\}^*\}$ ⟶ CFL But Not DCFL

PUSH POP PUSH POP

$L = \underbrace{ww^R}_{CFL} \cdot \underbrace{xx^R}_{CFL} = CFL$

$L = \underbrace{ww^R}_{A} \underbrace{xx^R}_{B}$

$S \longrightarrow AB$

$A \longrightarrow aAa \mid bAb \mid \epsilon$

$B \longrightarrow aBa \mid bBb \mid \epsilon$

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$$L = \underbrace{ww^R}_{A}\underbrace{xx^R}_{A}$$

$$L \neq ww^R.ww^R$$

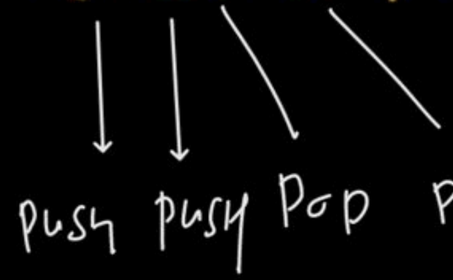
$$L = ww^Rxx^R$$

$$S \longrightarrow AA$$

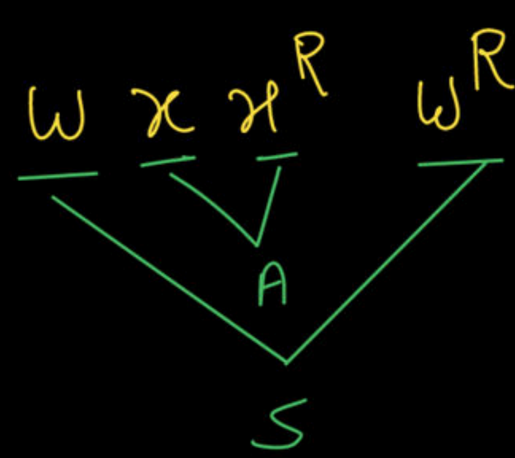
$$A \longrightarrow aAa \mid bAb \mid \epsilon$$

98. $L = \{wxx^R w^R \mid w, x \in \{a, b\}^*\}$ ⟶ CFL But Not DCFL

I



II $L = \{ w \; x \; x^R \; w^R \}$



$S \longrightarrow aSa \mid bSb \mid A$

$A \longrightarrow aAa \mid bAb \mid \in$

III

$L = \dfrac{wx}{w} \Big| \dfrac{x^R w^R}{w^R}$

Assume $wx = xw$ then it Becomes ww^R which is well know CFL

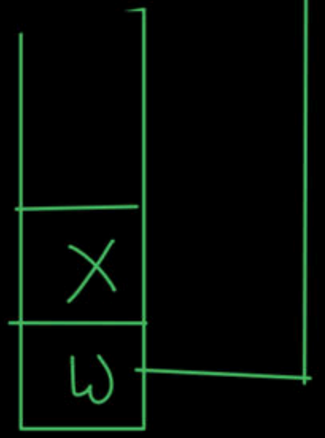
$(w \cdot x)^R = x^R w^R$

$(w \cdot x)^R \neq w^R x^R$

99. $L = \{wxw^R x^R | w, x \in \{a, b\}^*\}$ ⟶ CSL But Not CFL

$L = \{ w x w^R x^R \mid w \in (a,b)^* \}$

PUSH PUSH



can't compare with Bottom of stack

100. $L = \{a^n b^n \mid n \geq 0\} \cup \{a^n b^{3n} \mid n \geq 0 \cup \{a^n b^{5n} \mid n \geq 0\}$ ⟶ CFL But Not DCFL

Push 1a

Push 3a

Push 5a

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CFL but not DCFL

1. $L = \{a^n b^n \mid n \geq 0\} \cup \{a^n b^{2n} \mid n \geq 0\}$
2. $L = \{a^n b^m \mid m=n \text{ or } m=2n\}$ = 1
3. $L = \{a^n b^m \mid m=n \text{ or } m=2n \text{ or } m=3n\}$ = $a^n b^n \cup a^n b^{2n} \cup a^n b^{3n}$
4. $L = \{a^n b^n \mid n \geq 0\} \cup \{a^n b^{3n} \mid n \geq 0 \cup \{a^n b^{5n} \mid n \geq 0\}$
5. $L = \{a^m b^n c^p \mid m=n \text{ or } n=p$
6. $L = \{a^m b^n c^p \mid m=p \text{ or } n=p$
7. $L = \{a^m b^n c^p \mid m=p \text{ or } m=n$
8. $L = \{a^m b^n c^p d^q \mid m = p \text{ or } n = q$
9. $L = \{a^m b^n \mid m \leq n \leq 2m$
10. $L = \{a^m b^n \mid n \leq m \leq 2n$

11. $L = \{ww^R \mid w \in (a, b)^*$ or Even Length Palindrome

12. $L = wxw^R \mid w \in (a, b)^*, X \in (a, b)$ or Odd Length Palindrome

13. $L = wxw^R \mid w \in (a, b)^*, X \in (a, b, \epsilon)$ or All Palindrome

14. $L = wc^m w^R \mid w \in (a, b)^* \; m \geq 0$

15. $L = wcw^R \mid w \in (a, b, c)^*$

16. $L = wc^m w^R \mid w \in (a, b, c)^* \; m \geq 1$

17. $L = \{w\, a^n\, b^n \mid w \in (a, b)^* \; n \geq 1\}$

18. $L = \{ww^R\, a^n\, b^n \mid w \in (a, b)^* \; n \geq 0\}$

19. $L = \{ww^R \# a^n\, b^n \mid w \in (a, b)^* \; n \geq 0\}$

20. $L = \{wa^n b^n w^R \mid w \in \{a, b\}^*, n \geq 0\}$

21. $L = \{ww^R xx^R \mid w, x \in \{a, b\}^*\}$

22. $L = \{wxx^R w^R \mid w, x \in \{a, b\}^*\}$

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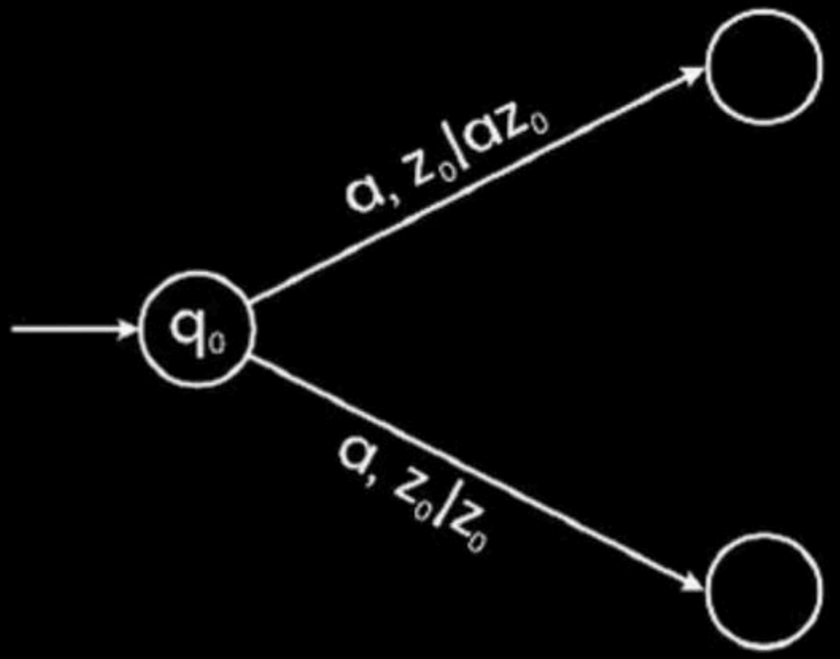
8. $L = \{a^m b^n c^p d^q \mid m = p \text{ OR } n = q\}$

$m = p$ **OR** $n = q$

$a^m \; b^n \; c^m \; d^q$ $\qquad$ $a^m \; b^n \; c^p \; d^n$

$a^m \; b^* \; c^m \; d^*$ $\qquad$ $a^* \; b^n \; c^* \; d^n$

PUSH $\qquad$ **SKIP**



NPDA But Not DPDA

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9. $L = \{a^m b^n \mid m \leq n \leq 2m\}$

NPDA But Not DPDA

CFL But Not DCFL

m	n	String	
1	1, 2	ab	
		abb	
2	2, 3, 4	aabb	
		aabbb	
		aabbbb	
3	3, 4, 5, 6	aaabbb	→ a → Push 1 a
		aaabbbb	
		aaabbbbb	
		aaabbbbbb	→ a → Push 2 a

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aaabbb

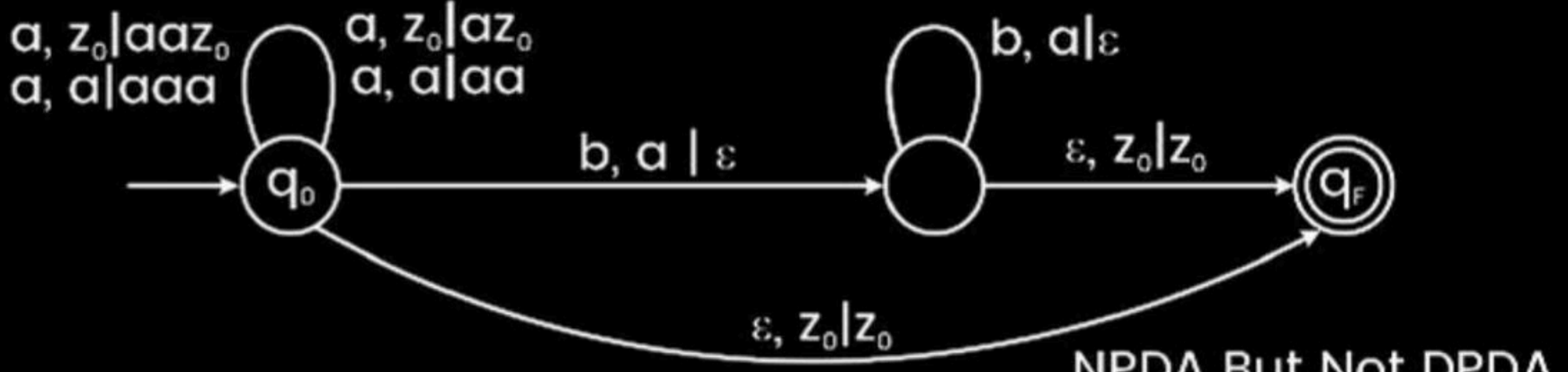
↑↑↑↑↑↑

a
a
a
z_0

aaabbbbbb

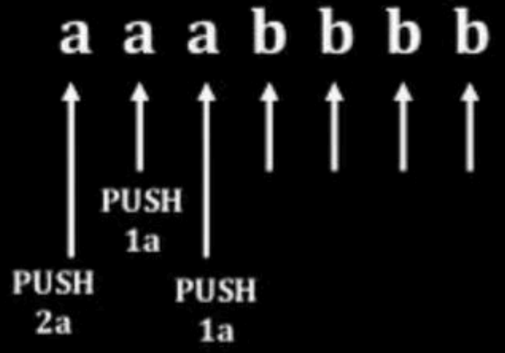
↑↑↑↑↑↑↑↑↑

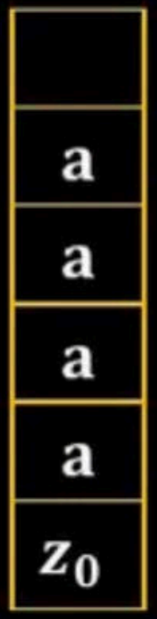
a
a
a
a
a
a
z_0

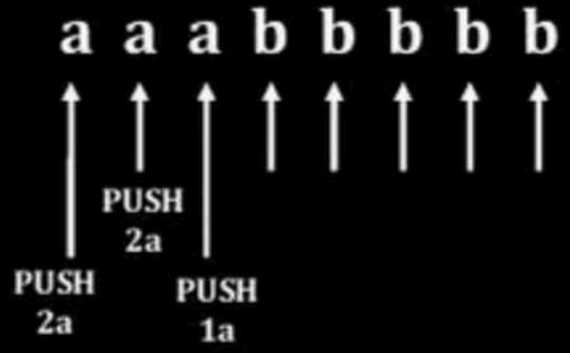


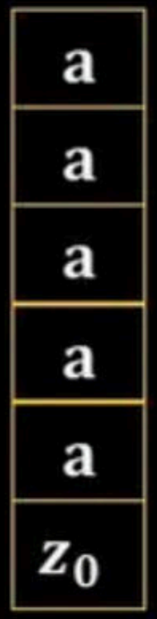
NPDA But Not DPDA

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a a a b b b b
PUSH 2a
PUSH 1a
PUSH 1a


a
a
a
a
a
z_0


a a a b b b b b
PUSH 2a
PUSH 2a
PUSH 1a


a
a
a
a
a
z_0

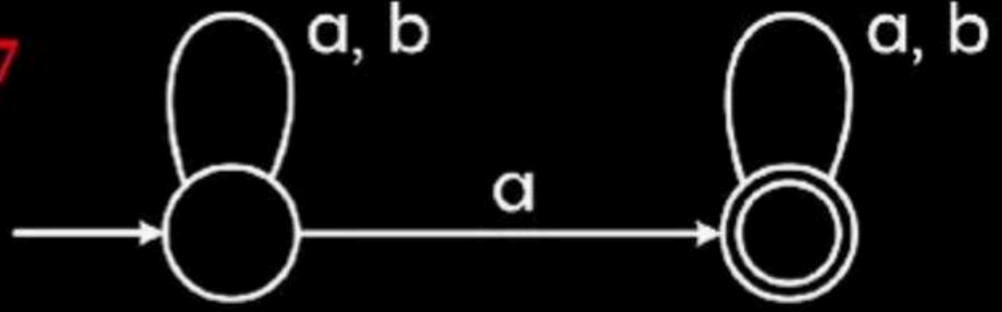
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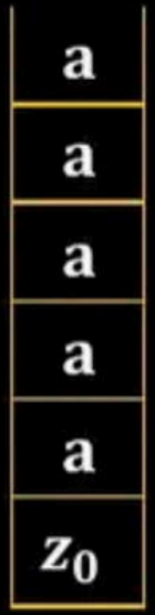
a a a b b b b → 3 4 2 5 6 6 6 7

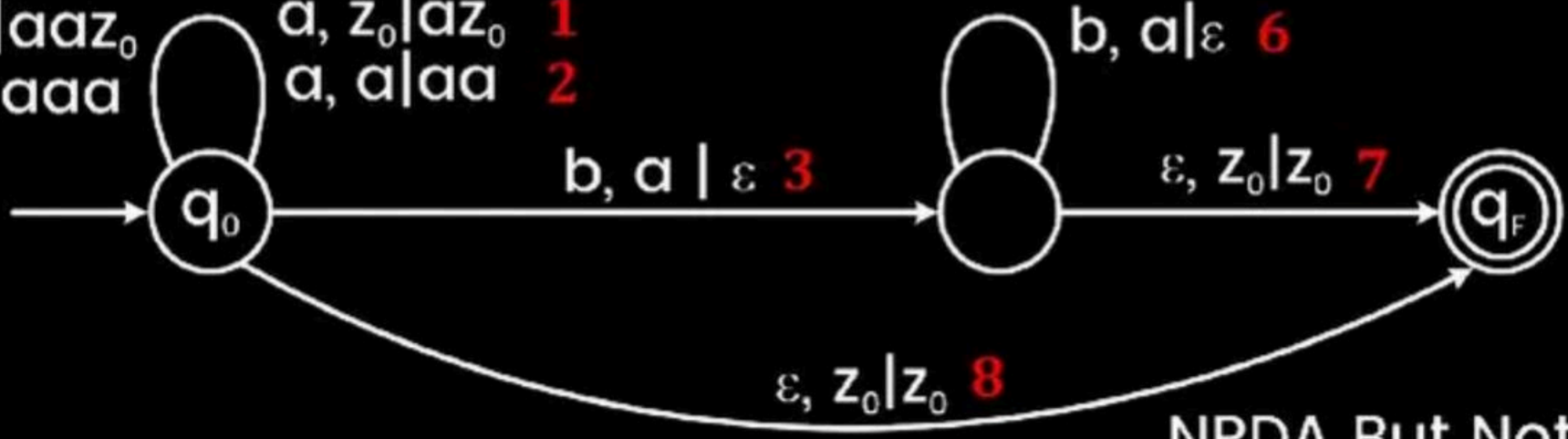
PUSH 2a

PUSH 2a

PUSH 1a



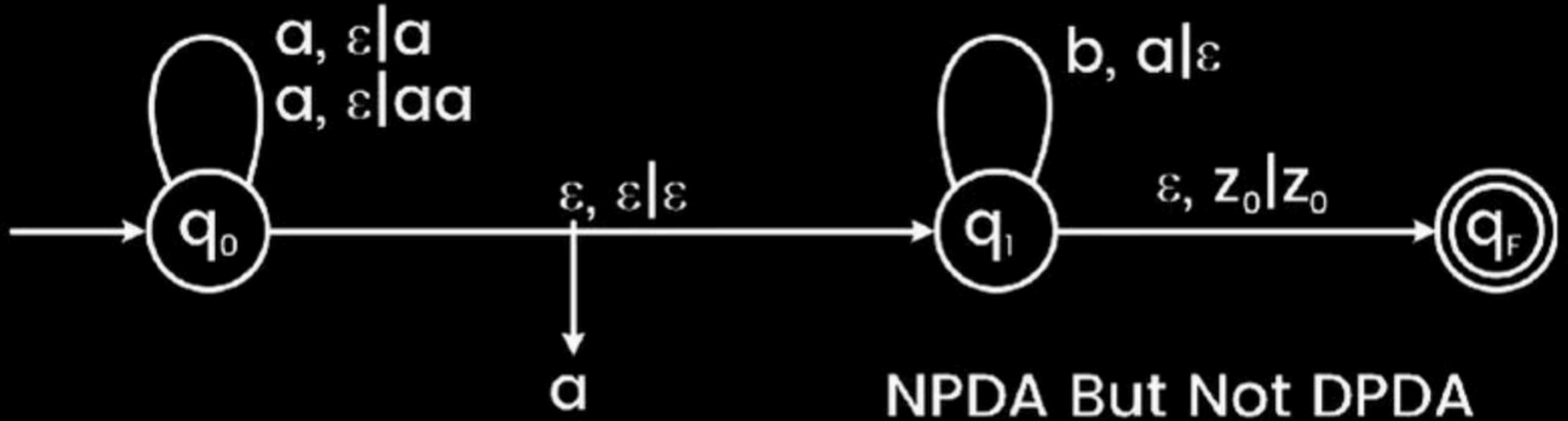




NPDA But Not DPDA

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OR



NPDA But Not DPDA

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10. $L = \{a^m b^n \mid n \leq m \leq 2n$

→ **NPDA But Not DPDA**

CFL But Not DCFL

n	m	String
1	1, 2	ab aab
2	2, 3, 4	aabb aaabb aaaabb
3	3, 4, 5, 6	aaabbb aaaabbb aaaaabbb aaaaaabbb

→ **For every a → Push 1 a**

For every b → Push 1 a

OR

PoP 2a

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CSL but Not CFL

1. $L = \{a^n b^n c^n \mid n \geq 0$
2. $L = \{a^n b^{2n} c^{3n} \mid n \geq 0$
3. $L = \{a^n b a^{2n} b a^{3n} \mid n \geq 0$
4. $L = \{a^m b^n c^p \mid m > n > p$
5. $L = \{ a^n b^m c^n d^m \mid m, n \geq 1$
6. $L = \{a^n b^n a^n b^n \mid n \geq 0$
7. $L = \{a^m b^n a^m b^n \mid n \geq 0$

CSL But Not CFL

8. $L = \{w \mid w \in (a, b, c)^* \text{ and } n_a(w) = n_b(w) = n_c(w)$
9. $L = \{w \mid w \in (a, b, c)^* \text{ and } n_a(w) > n_b(w) > n_c(w)$
10. $L = \{w \mid w \in (a, b, c)^* \text{ and } n_a(w) \leq n_b(w) \leq n_c(w)$
11. $L = \{w \mid w \in (a, b, c)^* \text{ and } n_a(w) \geqq n_b(w) \geqq n_c(w)$

CSL But Not CFL

12. $L = \{a^n b^m c^m \mid n > m\}$
13. $L = \{a^n b^m c^m \mid n < m\}$
14. $L = \{a^n b^m c^m \mid n \geq m\}$
15. $L = \{a^n b^m c^m \mid n \leq m\}$
16. $L = \{a^n b^m c^m \mid n = m\}$
17. $L = \{a^n b^m c^m \mid n \neq m\}$

CSL But Not CFL

12. $a^n b^m c^m \mid n > m$

18. $L = \{a^m b^n c^p \mid m=n \text{ and } n=p\} \longrightarrow a^m b^m c^m \mid m \geq 0$ or

19. $L = \{a^m b^n c^p \mid m=p \text{ and } n=p\}$ $a^m b^m c^m$

20. $L = \{a^m b^n c^p \mid m=p \text{ and } m=n\}$ $a^m b^m c^m$

$a^m b^n c^p \mid m=n=p$

↓

CSL

21. $L = \{a^m b^n c^p \mid p = m \times n$

22. $L = \{a^m b^n c^p \mid p > m \times n$

23. $L = \{a^m b^n c^p \mid p \geq m \times n$

24. $L = \{a^m b^n c^p \mid p < m \times n$

25. $L = \{a^m b^n c^p \mid p \leq m \times n$

26. $L = \{a^m b^n c^p \mid p \neq m \times n$

CSL But Not CFL

$L = a^m b^n c^{m \times n}$

↓

CSL

↓

PDA can't do multiplication, Division,

2^n, n^2, a^{prime}, $n!$

27. $L = \{a^m b^n c^p \mid p = m \mid n$

28. $L = \{a^m b^n c^p \mid p > m \mid n$

29. $L = \{a^m b^n c^p \mid p \geq m \mid n$

30. $L = \{a^m b^n c^p \mid p < m \mid n$

31. $L = \{a^m b^n c^p \mid p \leq m \mid n$

32. $L = \{a^m b^n c^p \mid p \neq m \mid n$

$L = a^m b^n c^{m/n}$

CSL But Not CFL

33. $L = \{a^n b^m c^{n \times m} \mid n, m \geq 0$

34. $L = \{a^{n/m} b^n c^m \mid n, m \geq 0$

35. $L = \{a^n b^{n \times m} c^m \mid n, m \geq 0$

CSL But Not CFL

36. $L = \{a^n b^m \mid m = n^2$ }

$L = a^n b^{n^2}$ → Not Regular → Not CFL

37. $L = a^m b^n c^p d^q \mid m \times n = p \times q$

38. $L = a^m b^n c^p d^q \mid m \times q = n \times p$

39. $L = a^m b^n c^p d^q \mid m \times p = n \times q$

CSL But Not CFL

40. $L = a^m b^n c^p d^q \mid m \mid n = p \mid q$

41. $L = a^m b^n c^p d^q \mid m \mid q = n \mid p$

42. $L = a^m b^n c^p d^q \mid m \mid p = n \mid q$

CSL But Not CFL

43. $L = \{ww \mid w \in (a, b)^*$

44. $L = \{www \mid w \in (a, b)^*$

45. $L = \{wcw \mid w \in (a, b)^*$

46. $L = \{ww^Rw^R \mid w \in (a, b)^*$

47. $L = \{w^Rww^R \mid w \in (a, b)^*$

48. $L = \{w^Rw^Rw \mid w \in (a, b)^*$

49. $L = \{ x\#y \mid x,y \in (a, b)^* \text{ and } x = y$

→ Copy Lang or Forward matching

CSL But Not CFL

$L = \{ X\#X \mid X \in (a,b)^* \}$

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Home work

1. $L = \{w\overline{w} \mid w \in (a, b)^*$
2. $L = \{w(\overline{w})^R \mid w \in (a, b)^*$
3. $L = \{ x\#y \mid x,y \in (a, b)^* \text{ and } x \neq y$

Note

unacademy

$L = \{ww \mid w \in (a, b)^*$ is not CFL but complement of ww is CFL

GATE CSE 1988

NAT

What is the type of the language L, where $L = \{a^n b^n \mid 0 < n < 327\text{-th prime number}\}$



Consider the languages:

- $L_1 = \{ww^R | w \in \{0,1\}^*\}$
- $L_2 = \{w \neq w^R | w \in \{0,1\}^*\}$, where $\neq$ is a special symbol
- $L_3 = \{ww | w \in \{0,1\}^*\}$

Which one of the following is TRUE?

A L_1 is a deterministic CFL

B L_2 is a deterministic CFL

C L_2 is a CFL, but not a deterministic CFL

D L_3 is a deterministic CFL

GATE CSE 2015

MCQ

Which of the following languages are context-free?

$L_1: \{a^m b^n a^n b^m | m, n \geq 1\}$

$L_2: \{a^m b^n a^m b^n | m, n \geq 1\}$

$L_3: \{a^m b^n | m = 2n + 1\}$

A L_1 and L_2 only

B L_1 and L_3 only

C L_2 and L_3 only

D L_3 only

GATE CSE 2016

MCQ

Consider the following languages:

- $L_1 = \{a^n b^m c^{n+m} : m, n \geq 1\}$
- $L_2 = \{a^n b^n c^{2n} : n \geq 1\}$

Which one of the following is TRUE ?

A Both L_1 and L_2 are context-free.

B L_1 is context-free while L_2 is not context-free.

C L_2 is context-free while L_1 is not context-free.

D Neither L_1 nor L_2 is context-free.

GATE CSE 1997

MCQ

Which of the following languages over $\{a, b, c\}$ is accepted by a deterministic pushdown automata ?

A $\{wcw^R | w \in \{a, b\}^*\}$

B $\{ww^R | w \in \{a, b, c\}^*\}$

C $\{a^n b^n c^n | n \geq 0\}$

D $\{w | w \text{ is a palindrome over } \{a, b, c\}\}$

Note: w^R is the string obtained by reversing' w'.

GATE CSE 2025

MCQ

Consider the following languages:

- $L_1 = \{a^m b^m c^{n+m} \mid m, n \geq 1\}$
- $L_2 = \{a^m b^n c^{n+m} \mid m, n \geq 1\}$

Which one of the following is statement is correct ?

A Both L_1 and L_2 are context-free Languages

B L_1 is context-free Languages but L_2 is not a context-free Language.

C L_1 is not context-free Language but L_2 is a context-free Language

D Neither L_1 nor L_2 is context-free Language.

GATE CSE 2004

MCQ

The language $\{a^m b^n c^{m+n} \mid m, n \geq 1\}$ is

A regular

B context-free but not regular

C context-sensitive but not context free

D type-0 but not context sensitive

GATE CSE 2008

MCQ

Which of the following is true for the language

$\{a^P \mid p \text{ is a prime}\}$?

A It is not accepted by a Turing Machine

B It is regular but not context-free

C It is context-free but not regular

D It is neither regular nor context-free, but accepted by a Turing machine

GATE CSE 2014

MCQ

Consider the following languages over the alphabet $\Sigma = \{0, 1, c\}$

$L_1 = \{0^n 1^n \mid n \geq 0\}$

$L_2 = \{wcw^r \mid w \in \{0, 1\}^*\}$

$L_3 = \{ww^r \mid w \in \{0, 1\}^*\}$

Here, w^r is the reverse of the string w. Which of these languages are deterministic Context-free languages ?

A None of the languages

B Only L_1

C Only L_1 and L_2

D All the three languages

GATE CSE 2010

MCQ

Consider the languages

$L1 = \{0^i1^j \mid i \neq j\}$,

$L2 = \{0^i1^j \mid i = j\}$,

$L3 = \{0^i1^j \mid i = 2j + 1\}$,

$L4 = \{0^i1^j \mid i \neq 2j\}$

A Only L2 is context free.

B Only L2 and L3 are context free.

C Only L1 and L2 are context free.

D All are context free

GATE CSE 2011

MCQ

Consider the languages L1, L2 and L3 as given below.

$L1 = \{0^p1^q \mid p, q \in N\}$,

$L2 = \{0^p1^q \mid p, q \in N \text{ and } p = q\}$ and,

$L3 = \{0^p1^q0^r \mid p, q, r \in N \text{ and } p = q = r\}$,

Which of the following statements is NOT TRUE ?

A Push Down Automata (PDA) can be used to recognize L1 and L2

B L1 is a regular language

C All the three languages are context free

D Turing machines can be used to recognize all the languages

GATE CSE 2006

MCQ

Which of the following languages is accepted by a non-deterministic pushdown automaton (PDA) but NOT be a deterministic PDA?

A $\{a^n b^n c^n | n \geq 0\}$

B $\{a^l b^m c^n | l \neq m \text{ or } m \neq n\}$

C $\{a^n b^n | n \geq 0\}$

D $\{a^m b^n | m, n \geq 0\}$

GATE CSE 2017

MCQ

Consider the following languages.

- $L_1 = \{a^p \mid p \text{ is a prime number}\}$
- $L_2 = \{a^n b^m c^{2m} \mid n \geq 0, m \geq 0\}$
- $L_3 = \{a^n b^n c^{2n} \mid n \geq 0\}$
- $L_4 = \{a^n b^n \mid n \geq 1\}$

Which of the following are CORRECT ?

I. L_1 is context free but not regular

II. L_2 is not context free

III. L_3 is not context free but recursive

IV. L_4 is deterministic context free

A I, II and IV only

B II and III only

C I and IV only

D III and IV only

GATE CSE 2006

MCQ

Let

- $L_1 = \{0^{n+m}1^n0^m | n, m \geq 0\}$
- $L_2 = \{0^{n+m}1^{n+m}0^m | n, m \geq 0\}$ and
- $L_3 = \{0^{n+m}1^{n+m}0^{n+m} | n, m \geq 0\}$

Which of these languages are NOT context free?

A L_1 only

B L_3 only

C L_1 and L_2

D L_2 and L_3

Let $\sum = \{a, b, c\ d\}$ and let $L = \{a^i b^j c^k d^l | i, j, k, l \geq 0\}$.

Which of the following constraints ensure (s) that the language L is context-free?

A $i + k = j + l$

B $i = k$ and $j = l$

C $i = l$ and $j = k$

D $i + j = k + l$

Consider the following languages:

I. $\{a^m b^n c^p d^q \mid m + p = n + q, \text{ where } m, n, p, q \geq 0\}$

II. $\{a^m b^n c^p d^q \mid m = n \text{ and } p = q, \text{ where } m, n, p, q \geq 0\}$

III. $\{a^m b^n c^p d^q \mid m = n = p \text{ and } p \neq q, \text{ where } m, n, p, q \geq 0\}$

IV. $\{a^m b^n c^p d^q \mid mn = p + q, \text{ where } m, n, p, q \geq 0\}$

Which of the above languages are context-free?

A I and IV only

B I and II only

C II and III only

D II and IV only

GATE CSE 2019

MCQ

Which one of the following languages over $\Sigma = \{a, b\}$ is NOT context-free?

A $\{ww^R | w \in \{a, b\}^*\}$

B $\{wa^n b^n w^R | w \in \{a, b\}^*, n \geq 0\}$

C $\{wa^n w^R b^n | w \in \{a, b\}^*, n \geq 0\}$

D $\{a^n b^i | i \in \{n, 3n, 5n\}, n \geq 0\}$

For a string w, we define w^R to be the reverse of w. For example, if $w = 01101$ then $w^R = 10110$.

Which of the following languages is/are context-free?

A $\{wxw^Rx^R | w, x \in \{0,1\}^*\}$

B $\{ww^Rxx^R | w, x \in \{0,1\}^*\}$

C $\{wxw^R | w, x \in \{0,1\}^*\}$

D $\{wxx^Rw^R | w, x \in \{0,1\}^*\}$

GATE CSE 2022

MSQ

Consider the following languages:

- $L_1 = \{ww | w \in \{a, b\}^*\}$
- $L_2 = \{a^n b^n c^m | m, n \geq 0\}$
- $L_3 = \{a^m b^n c^n | m, n \geq 0\}$

Which of the following statements is/are FALSE?

A L_1 is not context-free but L_2 and L_3 are deterministic context-free.

B Neither L_1 or L_2 is context-free.

C L_2, L_3 and $L_2 \cap L_3$ all are context-free.

D Neither L_1 nor its complement is context-free.

Consider the context-free grammars over the alphabet {a, b, c} given below. S and T are non-terminals.

$G_1 : S \rightarrow aSb \mid T, T \rightarrow cT \mid \epsilon$

$G_2 : S \rightarrow bSa \mid T, T \rightarrow cT \mid \epsilon$

The language $L(G_1) \cap L(G_2)$ is

A Finite

B Not finite but regular

C Context-free but not regular

D Recursive but not context-free

GATE CSE 2025

MCQ

Consider the two lists List I and list II

List I		List II	
(i)	Context Free Languages	(a)	Closed under union
(ii)	Recursive Languages	(b)	Not Closed under Complementation
(iii)	Regular Languages	(c)	Closed under Intersection

A (i) – (a), (ii) – (b), and (iii) – (c)

B (i) – (b), (ii) – (a), and (iii) – (c)

C (i) – (b), (ii) – (c), and (iii) – (a)

D (i) – (a), (ii) – (c), and (iii) – (b)

GATE CSE 2007

MCQ

Match the following:

E.	Checking that identifiers are declared before their use	P.	$L = \{a^n b^m c^n d^m \mid n \geq 1, m \geq 1\}$
F.	Number of formal parameters in the declaration of a function agrees with the number of actual parameters in a use of that function	Q.	$X \rightarrow XbX \mid XcX \mid dXf \mid g$
G.	Arithmetic expressions with matched pairs of parentheses	R.	$L = \{wcw \mid w \in (a \mid b)^*\}$
H.	Palindromes	S.	$X \rightarrow bXb \mid cXc \mid \epsilon$

A E-P, F-R, G-Q, H-S

B E-R, F-P, G-S, H-Q

C E-R, F-P, G-Q, H-S

D E-P, F-R, G-S, H-Q

Consider the following statement.

$L_1 = \{wxyx \mid w, x, y \in (0 + 1)^+\}$

$L_2 = \{xy \mid x, y \in (a + b)^*, |x| = |y|, x \neq y\}$

Which one of the following TRUE?

A L_1 is regular and L_2 is context-free.

B L_1 context-free but not regular and L_2 is context-free.

C Neither L_1 nor L_2 is context-free.

D L_1 context-free but L_2 is not context-free.

Let G = ({S}, {a, b}, R, S) be a context free grammar where the rule set R is

$S' \rightarrow aSb \mid SS \mid \epsilon$

Which of the following statements is true ?

A. G is not ambiguous

B. There exist x, y $\in$ L(G) such that xy $\notin$ L(G)

C. There is a deterministic pushdown automation that accepts L(G)

D. We can find a deterministic finite automaton that accepts L(G)

GATE CSE 1987

MCQ

FORTRAN is a:

A Regular language

B Context-free language

C Context-sensitive language

D None of the above

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THANK YOU!

Here’s to a cracking journey ahead!